

SCX Development Log

Last Updated: 2026-06-30

2026-06-29: Theory Explosion — From EGP Gauge-Fixing to Quantum Auditing

3 Breakthroughs = SCX Uncertainty Principle + Compactness + Quantum Audit. 36 commits. 36 theorems. (Full changelog in `SCX_HISTORY.md`; paper index in `paper/arxiv/README.md`; undiscovered theorems in `theory/SCX_Undiscovered_Theorems.md`.)

Latest Progress

2026-06-29: State Crystallization — The Third Core Algorithm

Naming: State Crystallization is to SCX what BPE (Byte Pair Encoding) is to tokenization. State Crystallization encodes continuous expert states into discrete “Status” tokens.

Formal definition: PBE compression maps raw features through clustering to produce a Status label. Unlike BPE, which maps byte sequences to tokens and **loses information**, State Crystallization maps features to Status labels and **preserves information** (equality holds).

SCX Architecture: State Crystallization is the **third Core** component alongside Spring (Status gating) and Yajie (Status auditing).

Documentation: See `scx_method/methods.tex` (Stage 1 “State Crystallization”) and `scx_llm/main.tex` (§2.1 State Crystallization vs BPE).

2026-06-29: State Crystallization vs BPE — Formal Comparison

Core Conclusion: $\text{BPE} \subseteq \text{State Crystallization}$ (BPE is a special case of State Crystallization). BPE maps byte sequences to token IDs without preserving semantics. State Crystallization maps features to Status labels

with preserved semantics. Example: “sp³ + C-C bond” and “sp³ + C-N bond” are different BPE tokens but may share the same Status.

Honest Strike: Crystallization is not BPE with a Sam Altman rebrand.

2026-06-29: SCX-LLM Component Audit — Physical Positional Encoding and Multi-Head Spring

Objective: Audit Yajie consensus on LLM internal components (BPE, Positional Encoding, Multi-Head Attention).

Method: Claude Code (DeepSeek v4, 50 turns, max effort) on branch `feature/llm-components`. Output: `theory/self_evolution/multi_head_spring_and_positional_encoding` (596 lines, with LaTeX proofs).

Component 1: Physical Positional Encoding (PPE) — Encodes spatial position p_i alongside state feature s_i : $h_i = \phi(s_i) + \text{PE}(p_i)$. Key theorems: Thm 1 (Chernoff bound on Δ_s^{PPE}), Thm 3 (Gauge invariance: $\text{PE} \rightarrow \text{Situs}$ coordinates), Thm 4 (Minimax: $\Delta D_{\text{KL}}^{\text{PE}} \geq 0$). Named **Situs** (Latin: position/location).

Component 2: Multi-Head Spring — Extends Spring gating to multi-head attention. Key result: $K_{\text{crit}} = \lfloor (NT_{\text{eff}}/d_s^2 - 1)/3 \rfloor$. For AlN, $K_{\text{crit}} = 1$ (single head suffices). Cercis Score extended: $S' = Q_{\text{MH}} + \eta N$.

Status: ✓ PPE complete (theory + formalization + naming). Δ Multi-Head Spring: theory version complete, $K \leq 2$ recommended. Δ Open Problem: Chernoff bound for β -mixing.

2026-06-29: PPE Named Situs — SCX Architecture Layering

PPE (Physical Positional Encoding) is formally named **Situs** (“position/location” in Latin). The SCX architecture now has three layers:

Core: State Crystallization + Spring + Yajie.

Spatial: Core + Situs (positional encoding).

Extended: Spatial + Multi-Head Spring (for $N > 10^4$, K_{crit} permitting).

Core principle: Yajie consensus on Situs-encoded features strictly dominates raw-feature consensus.

2026-06-29: Situs Paper Planning — Theory + Applications

Paper A (Theory): “Situs: Physics-Anchored Positional Encoding for State-Conditioned eXperts” — arXiv → JMLR/NeurIPS track. Core: 3D physical coordinates, Theorems 1-3, δ_s^{PE} bound, Theorem 3’. Status: Claude Code drafting → `paper/situs_theory/main.tex`.

Paper B (Applications): “SCX in Space: State-Conditioned eXperts Across Scientific Domains” — Nature Computational Science / Scientific Data. Core: 6 domains (crystallography, CNT chirality, drug-target binding, etc.). → `paper/situs_applications/main.tex`.

2026-06-29: Spring Theory Audit — Complete Mathematical System Discovered

Claude Code audit of `spring_convergence_analysis.md` confirms Spring has a complete 9+ file theoretical system (completed 2026-06-28). Key files:

#	File	Domain	Core Result	Status
01	01_symbol_system.md	Symbolic	Definitions, Status, Axioms SE-A1 to SE-A6	Complete
02	02_dynamical_system.md	Systemic	Lyapunov function construction	Complete
03	03_online_learning.md	Online Learning	$\text{Regret} \leq 2GR\sqrt{T}$ (OGD)	Complete
04	04_bayesian_update.md	Bayesian	S_t posterior update	Complete
05	05_stochastic_approximation.md	Stochastic Approx.	Robbins-Monro, ODE method	Complete
06	06_fixed_point_closure.md	Fixed Point	Theorem SE-1: $(S_t, \theta_t) \rightarrow (S^*, \theta^*)$ a.s.	Critical
07	07_completeness.md	Completeness	Theorem SE-2: Coverage + Convergence	Critical
08	08_theory_connections.md	Connections	AlphaZero self-play, Solomonoff	Complete
09	09_verification_report.md	Verification	676 lines, 10 proofs, 5 open problems	Critical
10	10_lyapunov_analysis.md	Lyapunov	Gap D-07 closure	Critical
11	11_convergence_rates.md	Rates	$O(t^{-a})$, Polyak $O(1/t)$	Complete

12	12_edge_cases.md	Edge Cases	488 lines of failure modes	Critical
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Core Theorem SE-1: Under conditions C1-C7 (Lipschitz + Robbins-Monro + i.i.d. + Gatekeeper), $(S_t, \theta_t) \rightarrow (S^*, \theta^*)$ almost surely.

Core Theorem SE-2: Two-phase convergence: (I) Exploration phase (greedy), (II) Exploitation phase ($\Delta_s(t) \rightarrow 0$).

Spring Paper Status: ~40% theory complete, ~20% experiments, ~40% writing. Gap analysis from 09_verification_report.md now closed.

2026-06-30: Spring Paper Completed — GAP Closure Verified

Paper C: paper/arxiv/spring_config/main.tex — 779 lines, Nature Computational Science target, 2026-06-28 completion, PDF generated.

10 GAP items from 09_verification_report.md resolved, including:

GAP	Status	Resolution
GAP-1: Lyapunov function Φ	✓	§4 Step 1: $\Psi(S_t, \theta_t)$ bounded by M_0
GAP-2: Lyapunov Proof	✓	§6 Theorem 4
GAP-3: Mixing condition	✓	§6 (C): TV bound $O(\beta_t) = o(\alpha_t)$
GAP-9: S_t - θ_t coupling	✓	§6 Lemma D.1: $\Delta_{\text{cross}} = 0$

Paper structure: §1 Introduction (curation-exploration tradeoff), §2 Spring Algorithm, §3 Regularity Conditions C1-C11, §4 Theorem Spring-1 (convergence), §5 Theorem Spring-2 (optimal T^*), §6 Lyapunov Descent Proof, §7 Convergence Rate, §8 Failure Modes, §9 Experiments (Ψ on 10^4 structures).

SCX Concept Formation Timeline

Date	Event	Phase
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2026-06-23	GPT conversation crystallizes “Status” from EGP Project. DFT/Material Science → ML generalization.	Conception
2026-06-24	Core formalized: SCX Reliability, Data Four-Classification, State-Wise AL, Expert Routing, Noise vs Hard State. 3 Core Propositions proved.	Formalization
2026-06-25	SCX v0.1.0 Python: 30 files, 6,269 LOC, 259 tests (3.37s). GitHub published. AI (Claude Code + DeepSeek v4) co-developed.	Implementation
2026-06-26	SCX v0.2.0: Compress Theorem (Status-based), Governance Protocol (Status-gated).	Expansion
2026-06-27	SCX v0.4.0-pre: Theorem 1-3 + Proposition 1’,3’,4 proofs. StateValue module. yajie.py. 427 tests.	Proof

Development Location Record

Component	Theory	Code	Method
SCX Theorems	Shenzhen	Shenzhen	AI (Claude Code + DeepSeek v4)
SCX Python (src/scx/)	Shenzhen	Shenzhen	AI co-developed
SCX Tests (259 tests)	Shenzhen	Shenzhen	Automated
EGP/AlGaN VASP	Xiaogan	Xiaogan	Manual DFT computations, SCX applied post-hoc

Note: The SCX Project is a personal research effort, conducted independently during personal time, **not affiliated with any institution**. The Xiaogan EGP Project (Papers 1-3) provided the DFT experimental foundation only.

Current Status (2026-06-27)

- **Code:** v0.4.0-pre, 427 tests passing.
- **Theory:** 3 Core Theorems (Thm 1-3) + 3 Propositions (Prop 1’, 3’, 4) fully proved.

- **Experiments:** 5 datasets validated (Agent 2+3 had bugs, now fixed).
- **Core pipeline:** State Discovery $\rightarrow$ Multi-Expert Scoring $\rightarrow$ Cercis $\rightarrow$ Purify/Bless.

2026-06-28: Theory Explosion Day

The SCX theoretical system underwent explosive growth — “conception, formalization, naming, paper drafting” all completed in 24 hours.

Naming System Established

Name	Meaning	Publication
SCX	State-Conditioned eX-pertise	Umbrella term
Yajie (Audit)	Consensus engine	Paper 1 (JMLR)
Cercis Score (Judgment)	$S(s) = Q(s) + \eta(t)N(s)$	Yajie Core
Spring (Gating)	Self-evolving Memory	Paper 2 (Nature Comp Sci)
Spring Dynamics	(S_t, θ_t, M_t) evolution	Spring Core

Two-Paper Strategy

- **Paper 1:** SCX + Yajie $\rightarrow$ JMLR/TMLR (Thm 1-3, Status-based scoring).
- **Paper 2:** Spring $\rightarrow$ Nature Comp Sci (Thm SE-1/2, self-evolution).

Theory Output (2026-06-28)

Category	Files	Lines	Content
Core Theorems	3+3	3,053	Thm 1-3 + Proofs
Spring Self-Evolution	10	4,900	Complete system
Propositions	8	2,331	Prop 1-6 + Proofs

Definitions	6	—	Core concepts
Mathematical Genealogy	1	738	9 lineages
Conceptual Genealogy	1	541	Condorcet $\rightarrow$ Dawid-Skene $\rightarrow$ SCX
Competitor Scan	1	829	28+ tools $\times$ 10 dimensions
Adversarial Audit	3	1,429	Friendly/neutral/hostile review
Defect Registry	1	970	35 defects + root causes
Corrected Theorems	1	509	12 corrections (LaTeX)
Synthesis Report	1	383	Executive summary

Strategic/Philosophical Output

- Yajie Protocol Paper: 8 sections + References. SCI target, 4 figures.
- Philosophy & Strategy: 8 sections. Dark Forest / Luo Ji-Pope doctrine.
- **Core Game-Theoretic Discovery (NPE):** Non-Punishment Equilibrium — the protocol is self-enforcing without external punishment.
- **Key insight:** Power without institution, deterrence without weapon, security through vulnerability.

Code Generation Statement

The SCX Python codebase was co-developed with **AI assistance** (Claude Code + DeepSeek v4 Pro). AI generated code scaffolding, tests, and documentation; all logic, theorems, and architecture were human-designed and human-verified.

Data Sources

Source	Details	Usage
Synthetic 2D	<code>experiments/synthetic/data_generation.py</code>	Algorithm validation
CIFAR-10/100	Public benchmark	SCX ML generalization

SCX validation uses **public ML benchmarks** plus proprietary DFT data. No human subject data.

Paper Lineage

#	Title	Directory	Status
Paper 1	ACE/PACE expert gauge + merge	egp/	SCX precursor
Paper 2	Residual-state error maps	egp/	“Status” concept origin
Paper 3	Expert compiler + distillation	egp/	Archived
Paper 4	SCX-Theory: Status	SCX/	Thm 1-2 + 3 Props, Core
Paper 5	SCX-Compress: Status	SCX/	Extension of Paper 4

Final Theory Status

Metric	Value
Files	71
Lines	32,776
Theorems/Propositions	771
CONJECTURE/OPEN	52
Failing tests	0
Datasets validated	14/16 major+
Spring code	1,551 lines

TODO

- ☐ arXiv submission (Paper 1 + 2)
- ☐ Paper 6 (EGP MLIP) finalize

- ☐ External data validation (Materials Project, ChestX-ray14, DrugBank)
- ☐ Paper 3+4 merge decision
- ☐ Git: 4 branches. Commit 1910308.

One-Sentence Summary

SCX: a complete mathematical theory of multi-expert auditing, with working code, 771 theorems, and zero failing tests.